CERTAIN-COAST — brief for a partner-country team

Certified early warning to protect water, port and energy infrastructure on a sinking Adriatic coast

NATO Science for Peace and Security (SPS) Programme · Multi-Year Project · call 2027-1 · submission by 15 November 2026, 23:59 CET · coordinated by the Faculty of Geology and Mining, Polytechnic University of Tirana, Albania

The project in brief

Near Durrës, the main port of Albania, the wellfields that supply drinking water, the port and an energy landfall stand on soft ground that is slowly sinking, close to active faults (the Mw 6.4 earthquake of 26 November 2019 struck there). Three hazards threaten them together: seawater entering the wells, liquefaction in an earthquake, and coastal flooding in storms. Sinking land and rising seas worsen all three. CERTAIN-COAST builds one model of the three hazards, a fast machine-learned copy of it whose error is bounded for every forecast, and warnings that state in advance how often they may miss a dangerous event. The warnings run live for twelve months with the water utility, the port, the energy operator and civil protection, and are then handed over to them. **36 months; about €382,500 requested from NATO.**

Why it fits the call

The 2027-1 call names among its topics of particular interest "protection of critical infrastructure, supplies and personnel; maritime and undersea infrastructure resilience". CERTAIN-COAST protects the water, port and energy infrastructure of a NATO Ally, and the method transfers to other subsiding coasts of Allies and partners, which is where the partner country comes in.

What is new

- **Warnings with a stated miss rate**, verified during live operation with a test that stays valid however often the record is checked.
- **A certified safe operating window**: the pumping and dewatering decisions that keep all three hazards within limits at once, and the sea level or year at which this window closes. In our literature search we found no published method for this.
- **The value of a sensor in years of safe operation**, used to choose the monitoring network.
- **Safe pumping tests chosen for the years they add** to the certified window, kept below a chloride test limit; in a synthetic test they raised the certified horizon from 10.8 to 31.9 years.

The role we propose for the partner-country team: clear tasks, no algorithm work

The coupled model, the machine-learning surrogate and the certified-alarm algorithm are built and run by the Albanian team (WP1–WP5, WP7). The partner team needs no specialist modelling staff. It leads **WP6, Training, transfer and dissemination**, with four concrete tasks:

Task	What the partner team does	When
Exchanges	Sends its early-career researchers to Durrës for two-month stays (six stays in total), where they work with the data, the operators and the Albanian team	Years 1–3
Transfer study	Collects public maps and data and local knowledge on a coast of its own country; the Albanian team runs the model and the algorithm with them; joint report and paper	Months 24–36
Regional workshop and outreach	Organises one workshop in its country, presents the results at conferences, and translates the public summary and the policy brief	Year 3
User-side review	Reads the bulletins and the operating procedures as an outside user and sends comments	Months 23–36

The Project Director also sits on the steering committee (online meetings are fine) and sends a short input to the four reports to NATO. The partner team co-authors the joint papers and keeps the open software.

Budget for the partner institution in the current plan: €69,000 (18% of the total)

Item	Partner share	Basis
Consortium meetings (kick-off, mid-term, final): travel and subsistence	€10,800	shared equally
Young-researcher exchanges, two months each	€21,000	partner
Conference presentations	€7,200	shared equally
Stipends for young researchers without a PhD: 4 researchers × 30 months × €400	€24,000	shared equally
Open-access publication fees	€6,000	shared equally

The share can be raised, for example with more exchanges or stipends at the partner institution; we will agree the final figures together before submission.

Who can be the partner-country Project Director

- Employed and resident in an eligible partner country of the SPS Programme, at a university, public research institute or other non-profit institution (companies cannot be co-directors).
- Holds no other SPS grant at the time (one SPS grant per person).
- Gives about 20% of working time to the project; co-signs the SPS Agreement for Joint Research with a representative of the institution.

What we need: everything by Thursday 5 November 2026

What	Details
Reply	Interest in principle, the name of the proposed Project Director and the institution
Partner form (attached, Word)	Details of the Project Director for the NATO application form, the partner team, the coast for the transfer study, and a one-page overview of the institution
Agreement for Joint Research (attached, NATO template)	Signed by the Project Director and by the Head of Institution and/or Financial Authority, scanned to PDF

The NATO-country Project Director then submits the application by 15 November 2026, 23:59 CET.

The full proposal text is shared as soon as you confirm interest. Contact: Dulian Zeqiraj, Faculty of Geology and Mining, Polytechnic University of Tirana (please reply to the e-mail that brought this brief).